

COST ACTIONS

COST Open Call Proposal

European Network on Urban Heat Resilience URBANHEAT

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Abstract

URBANHEAT is a four-year COST Action establishing the first pan-European, interdisciplinary and inclusive research and stakeholder network dedicated to urban heat resilience. Bringing together climate scientists, urban planners, public health researchers, social scientists and city practitioners from across COST Member Countries — with the majority of participants drawn from Inclusiveness Target Countries — the Action coordinates the production and exchange of knowledge on urban heat adaptation that no nationally funded project can generate in isolation. Organised around five Working Groups covering heat metrics, nature-based solutions, health and social vulnerability, governance, and dissemination, URBANHEAT harmonises methodologies, synthesises evidence, co-produces policy guidance with city and health authorities, and builds the capacity of early-career investigators. Its outputs include harmonised indicator frameworks, open and FAIR-compliant comparative datasets, peer-reviewed evidence syntheses, practitioner guidance documents, and Policy Briefs targeting EU institutions, national ministries and municipal authorities. Gender equity and open science principles are integrated structurally across all Working Groups. URBANHEAT directly supports the European Green Deal, the EU Adaptation Strategy and the EU Mission on Climate-Neutral and Smart Cities, and will leave a durable legacy of interdisciplinary researchers and shared knowledge infrastructure for urban heat adaptation across Europe.

Keywords: urban heat island, climate adaptation, nature-based solutions, public health, urban governance, social vulnerability, early-career researchers, interdisciplinary networking, inclusiveness target countries, European Research Area

SECTION 1 - SCIENCE AND TECHNOLOGY EXCELLENCE

1.1 Challenge and S&T Objectives

Urban heat stress constitutes one of the most urgent and multi-dimensional challenges confronting European cities in the context of accelerating climate change. Heat-related mortality, urban heat island intensification, compounding drought–heat events, and widening social vulnerability gaps are documented across the continent; yet the scientific and policy communities responding to these pressures remain fragmented along disciplinary, institutional and national lines. No pan-European coordination mechanism currently integrates climate science, urban planning, public health and social science in a sustained, structured and inclusive manner. URBANHEAT addresses this gap directly.

The primary objective of URBANHEAT is to establish a durable, inclusive and interdisciplinary research and stakeholder network that coordinates knowledge production and exchange on urban heat adaptation across COST Member Countries. The Action pursues five specific objectives: (1) to develop harmonised, interoperable metrics and indicators for measuring urban heat exposure, vulnerability and adaptation performance across diverse European urban typologies; (2) to synthesise evidence on the effectiveness and co-benefits of nature-based solutions (NbS) as urban heat mitigation tools, with explicit attention to equity and context-specificity; (3) to generate and disseminate consolidated knowledge on the health impacts of urban heat by integrating epidemiological, physiological and social-determinants perspectives; (4) to map and compare urban heat governance frameworks across COST Member Countries, identifying barriers to and enablers of effective policy uptake; and (5) to build the capacity of early-career investigators (ECIs), practitioners and policymakers — particularly those based in Inclusiveness Target Countries (ITCs) — to engage with and contribute to state-of-the-art urban heat research and governance.

These objectives are both ambitious and timely. They align directly with the European Green Deal, the EU Adaptation Strategy, the Mission on 100 Climate-Neutral and Smart Cities, and the EU4Health programme. Taken together, URBANHEAT will generate the shared conceptual frameworks, methodological standards, transnational datasets and human capital that nationally funded projects cannot produce in isolation. The Action thus represents a qualitatively new level of European coordination on urban heat resilience — one whose scientific and societal value is intrinsically dependent on its pan-European, interdisciplinary character.

1.2 Progress Beyond the State of the Art

The current body of knowledge on urban heat is substantial but structurally fragmented. Leading research groups in climate science, urban morphology, epidemiology, landscape ecology and urban governance are distributed across Europe and largely operate in parallel, each producing valuable but discipline-specific knowledge. Existing coordination efforts — including several past and current COST Actions in adjacent domains — have addressed energy efficiency in buildings, climate services or urban ecology in partial ways, but none has integrated the full socio-technical chain from heat exposure measurement through nature-based intervention to health outcome monitoring and governance response.

Urban heat island (UHI) research is maturing in its physical science dimensions, yet systematic comparative frameworks for vulnerability mapping remain underdeveloped. Nature-based solutions have gained prominent positions in EU policy discourse, but rigorous evidence syntheses on their heat mitigation performance under diverse urban and climatic conditions are scarce. Urban heat governance studies exist mostly as single-country case studies with limited comparative reach. URBANHEAT moves decisively beyond this state of affairs by providing, for the first time, a coordinated platform that: (a) harmonises metrics across disciplines and national contexts; (b)

enables large-scale comparative empirical work that is impossible within any single national research system; and (c) connects scientific knowledge production directly to municipal and regional decision-making processes through structured stakeholder engagement.

UHEAT is further distinguished from predecessor networks by its structural commitment to Inclusiveness Target Countries. Urban heat challenges are particularly acute in Southern and Eastern Europe — precisely the regions where ITC institutions are concentrated — and URBANHEAT is designed to leverage that geographic alignment to ensure that ITC researchers lead thematic contributions rather than participate peripherally. This design choice both strengthens the science and fulfils COST's core mandate of broadening participation in European research.

1.3 Scientific and Technological Approach

URBANHEAT is organised around five thematic Working Groups (WGs) that together span the full knowledge chain from measurement through intervention to governance. WG1 (Heat Metrics and Exposure Assessment) develops common indicator frameworks and data-sharing protocols. WG2 (Nature-Based Solutions for Urban Heat Mitigation) synthesises and evaluates green and blue infrastructure interventions. WG3 (Heat, Health and Social Vulnerability) integrates epidemiological, environmental health and social science perspectives. WG4 (Governance, Policy and Urban Planning) maps regulatory frameworks and co-produces policy guidance with municipal partners. WG5 (Dissemination, Capacity Building and Stakeholder Engagement) ensures that all knowledge outputs reach scientific, practitioner and policy audiences effectively.

The networking methodology draws on the full portfolio of COST instruments. Annual Management Committee (MC) meetings provide strategic oversight. WG meetings (two per year per WG) enable intensive scientific exchange and collaborative output development. Short-Term Scientific Missions (STSMs) facilitate researcher mobility, with priority given to ECIs and ITC-based researchers. Annual Training Schools — co-organised with ITC institutions — build methodological capacity for graduate students and early-career researchers. ITC Conference Grants and Young Researcher and Innovator (YRI) Conference Grants expand participation in international venues. Virtual Mobility grants supplement physical mobility for researchers who cannot travel. Dissemination and communication activities are embedded across all WGs and consolidated through WG5.

The Action operates on an open science model throughout: shared data protocols, pre-registration of collaborative studies, open-access publication of all Action outputs, and freely available training materials. Gender and social equity dimensions are integrated as explicit analytical categories across all WGs. Knowledge co-production with city networks and public health agencies is built into the mandates of WG3 and WG4 from the outset — not appended as an afterthought — ensuring that the pathway from scientific output to policy application is direct, traceable and accountable.

1.4 Interdisciplinary Considerations

URBANHEAT is intrinsically interdisciplinary. The challenge of urban heat adaptation cannot be addressed within any single discipline: physical measurement of heat exposure requires atmospheric science and remote sensing expertise; understanding effective mitigation requires landscape ecology, urban morphology and civil engineering; health impact assessment requires epidemiology, environmental health science and social medicine; and governance analysis requires political science, public administration and urban planning. The Action deliberately recruits proposers with expertise across all these domains and structures WG activity to require cross-disciplinary collaboration on joint outputs, including a formal requirement that all major deliverables carry co-authorship from researchers representing at least three distinct disciplines.

The Action also engages cross-sectoral partners: municipal authorities, regional health agencies, urban planning directorates, NGOs active in climate justice, and European city networks such as

Eurocities and the Covenant of Mayors. This trans-disciplinary dimension — integrating academic knowledge production with practitioner knowledge and lived policy needs — is central to the Action's theory of change. Joint working sessions between WG researchers and city partners are programmed within WG4 and form the foundation of the Action's policy uptake pathway.

Gender and social equity dimensions are integrated as analytical categories across all WGs and are not treated as optional additions. A dedicated Gender Balance Coordinator within the MC monitors representation in WG leadership and speaker slots at Action events, targeting a minimum of 40% representation for each gender. By embedding equity considerations at both the governance level and the substantive research agenda — including the analysis of socially differentiated heat vulnerability in WG3 — URBANHEAT ensures that its scientific and structural commitments to inclusion are mutually reinforcing.

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SECTION 2 - IMPACT

2.1 Expected Impact and Benefits

The primary scientific impact of URBANHEAT is the creation of shared methodological frameworks and comparative empirical knowledge that no single nationally funded project can produce. By harmonising heat exposure metrics (WG1), the Action enables systematic, cross-national comparison of urban heat vulnerability at European scale for the first time. By synthesising NbS evidence across climate zones and urban typologies (WG2), the Action provides the evidence base urgently needed by city planners and regional authorities. By connecting health outcomes to governance responses (WG3–WG4), the Action constructs the causal narratives necessary for credible policy advocacy.

The pathway to policy impact runs through the structured stakeholder engagement programme of WG4 and WG5. The Action will co-produce at least two Policy Briefs per year targeting EU institutions, national ministries and municipal authorities. It will engage directly with the EU Mission on Climate-Neutral and Smart Cities, the European Environment Agency's urban adaptation portal, and national adaptation platforms. Memoranda of Understanding with at least three European city networks will formalise institutional uptake channels. The long-term aim is to ensure that URBANHEAT outputs inform the next generation of EU and national urban climate adaptation plans, building codes incorporating heat resilience standards, and public health early-warning systems — providing a concrete and traceable line from Action outputs to regulatory and planning instruments.

Capacity building constitutes a sustained and independently significant impact pathway. Through Training Schools, STSMs and YRI grants, URBANHEAT will develop a cohort of interdisciplinary researchers — particularly from ITCs — capable of leading urban heat adaptation research and policy engagement independently after the Action concludes. This human capital effect is a durable legacy that extends well beyond the four-year Action period and contributes directly to the long-term strengthening of the European Research Area.

2.2 Dissemination and Stakeholder Engagement

URBANHEAT adopts a tiered dissemination strategy calibrated to distinct audiences. For the scientific community: joint peer-reviewed publications targeting high-impact journals in climate science, public health and urban studies; conference presentations at major European and international venues (EGU, EMS, ISEE, EURA); and open-access sharing of datasets and methodological protocols via established repositories. For practitioners and policymakers: Policy Briefs, a publicly accessible urban heat knowledge portal maintained by WG5, webinars for city networks, and contributions to national adaptation reporting processes. For the general public and urban communities: a communication plan led by WG5 including social media engagement, press releases coordinated with institutional communication offices, and participation in public engagement events such as European Researchers' Night.

All publications resulting from COST-funded networking activities will be made open-access. Data produced through collaborative Action activities will be deposited in FAIR-compliant repositories in line with COST and Horizon Europe open science expectations. The Action website, hosted by the coordinating institution and maintained throughout the four-year period, will serve as the central hub for all resources, training materials and stakeholder outputs. A dedicated presence on LinkedIn and X (formerly Twitter) will sustain ongoing communication with the urban policy community.

Inclusiveness is integral to the dissemination strategy: Training School materials will be made freely available online in accessible formats; all webinars will be recorded and captioned; and WG5 will monitor the geographic distribution of dissemination activities to ensure that ITC institutions are represented as co-authors, co-presenters and co-producers of policy outputs — not merely as

recipients of knowledge generated elsewhere. This monitoring function is reported annually to the Management Committee and feeds directly into strategic adjustments to the dissemination programme.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
Number of COST Member Countries represented in the Action network	At least 20 countries, with ITC countries comprising at least 50% of the network	Annual network census reported to Management Committee
Number of peer-reviewed joint publications co-authored across at least three COST countries	At least 30 over 4 years	Publication database maintained by Action Chair's institution
Number of ECIs (within 8 years of PhD) participating in Action activities	At least 40% of all Action participants	Annual participant survey and registration data
Number of STSMs completed, with priority to ITC and ECI researchers	At least 60 STSMs over 4 years, of which at least 60% involving ITC institutions	STSM application and completion records in e-COST
Number of Training Schools organised	At least 4, co-organised with ITC institutions in at least 3 cases	Training School reports submitted to Management Committee
Number of Policy Briefs and practitioner outputs delivered	At least 8 Policy Briefs; at least 2 formal engagements with city networks	WG5 dissemination log reviewed semi-annually
Gender balance in Action leadership and activities	At least 40% representation of each gender in WG leadership positions and speaker slots at Action events	Annual gender monitoring report by Action Gender Balance Coordinator

SECTION 3 - IMPLEMENTATION AND NETWORKING

3.1 Working Groups and Activities

WP	Title	Lead	PM	Duration
WP1	WG1 – Heat Metrics and Exposure Assessment	University of Athens (Greece)	0	M1-M48
WP2	WG2 – Nature-Based Solutions for Urban Heat Mitigation	ITC partner institution (Eastern/Southern Europe)	0	M3-M48
WP3	WG3 – Heat, Health and Social Vulnerability	ITC partner institution (Balkan region or Eastern Europe)	0	M3-M48
WP4	WG4 – Governance, Policy and Urban Planning	ITC partner institution (Central or Eastern Europe)	0	M6-M48
WP5	WG5 – Dissemination, Capacity Building and Stakeholder Engagement	Coordinating institution with ITC co-lead	0	M1-M48

WP1: WG1 – Heat Metrics and Exposure Assessment

Lead: University of Athens (Greece) | Person-months: 0 | Duration: M1 - M48

WG1 develops harmonised frameworks for measuring urban heat exposure and vulnerability across diverse European urban contexts. It conducts a systematic review of existing national and international metrics, identifies interoperability gaps, and produces consensus-based indicator protocols that all WG activities adopt as shared baselines. A data governance mapping exercise in Year 1 identifies national restrictions on data sharing and informs the design of the Action's open data infrastructure. WG1 outputs include a comparative indicator framework, a data-sharing protocol, and a living atlas of urban heat exposure across participating countries — all published in FAIR-compliant open repositories.

WG1.T1: Review and gap analysis of existing urban heat metrics - Systematic review of heat exposure and vulnerability indicators used across COST Member Countries, identifying methodological inconsistencies, interoperability barriers and national data governance constraints that affect cross-national data sharing.

WG1.T2: Development of harmonised indicator framework - Co-design of a consensus-based, interoperable urban heat indicator framework through iterative WG workshops and expert consultation, incorporating feedback from WG2, WG3 and WG4 to ensure cross-disciplinary applicability.

WG1.T3: Urban heat exposure atlas - Collaborative compilation and open publication of a comparative urban heat exposure dataset drawing on national data contributed by network members, with federated analysis approaches applied where full data sharing is legally constrained.

WP2: WG2 – Nature-Based Solutions for Urban Heat Mitigation

Lead: ITC partner institution (Eastern/Southern Europe) | Person-months: 0 | Duration: M3 - M48

WG2 synthesises evidence on the effectiveness of green and blue infrastructure interventions for urban heat mitigation across European climate zones and urban morphologies. It evaluates co-benefits, trade-offs, equity implications and implementation barriers through systematic reviews, meta-analyses and collaborative case study protocols. WG2 builds directly on the indicator framework developed by WG1 and coordinates with WG3 on health co-benefit assessment. Its principal output is practitioner-oriented NbS guidance aligned with EU Green Infrastructure policy and directly transferable to municipal planning contexts.

WG2.T1: Systematic evidence synthesis on NbS heat performance - Collaborative systematic review and meta-analysis of documented temperature reduction effects of urban greening, blue infrastructure and combined NbS interventions across European and comparable international contexts, following pre-registered review protocols.

WG2.T2: Context-sensitive NbS typology and decision support - Development of a climate-zone- and urban-morphology-sensitive NbS typology to guide practitioner selection of interventions, integrating equity analysis, co-benefit quantification and implementation barrier assessment into a structured decision-support framework.

WP3: WG3 – Heat, Health and Social Vulnerability

Lead: ITC partner institution (Balkan region or Eastern Europe) | Person-months: 0 | Duration: M3 - M48

WG3 integrates epidemiological, environmental health and social science expertise to generate consolidated knowledge on the health impacts of urban heat, with explicit attention to the social determinants of vulnerability. It reviews and harmonises heat-health surveillance methodologies, maps vulnerable population groups and develops cross-national comparative analyses linking heat exposure to health outcomes. WG3 coordinates closely with WG1 on shared exposure metrics and with WG4 on translating health evidence into policy-relevant governance recommendations. Gender and intersectional dimensions of heat vulnerability are treated as core analytical categories throughout.

WG3.T1: Harmonisation of heat-health surveillance methods - Review and alignment of epidemiological surveillance approaches and indicator definitions used across national health systems, producing a harmonised methodology protocol that enables rigorous cross-national comparison of heat-health outcomes.

WG3.T2: Vulnerability mapping and social determinants analysis - Collaborative mapping of socially vulnerable population groups — including older adults, low-income households, outdoor workers and migrants — and structural analysis of the social, spatial and policy determinants shaping differential heat health risk across participating countries.

WP4: WG4 – Governance, Policy and Urban Planning

Lead: ITC partner institution (Central or Eastern Europe) | Person-months: 0 | Duration: M6 - M48

WG4 maps and comparatively analyses urban heat governance frameworks, regulatory instruments and adaptation planning processes across COST Member Countries. It engages directly with municipal and regional authorities through co-production workshops, identifies implementation barriers, and develops transferable governance models grounded in evidence from WG1–WG3. WG4 is the primary interface between URBANHEAT and its city and policy stakeholders, and it anchors the Action's policy impact pathway through Memoranda of Understanding with at least three European city networks and a structured Policy Brief series.

WG4.T1: Comparative governance and policy mapping - Structured comparative analysis of national and municipal legal, planning and financial frameworks for urban heat adaptation across participating countries, identifying enabling conditions and systemic barriers to effective policy implementation.

WG4.T2: Stakeholder co-production workshops and policy guidance - Series of joint workshops with municipal authorities, regional health agencies and city networks to co-produce transferable policy guidance documents and governance recommendations, drawing directly on the evidence outputs of WG1–WG3.

WP5: WG5 – Dissemination, Capacity Building and Stakeholder Engagement

Lead: Coordinating institution with ITC co-lead | Person-months: 0 | Duration: M1 - M48

WG5 coordinates all dissemination, communication, Training School organisation, STSM and Virtual Mobility administration, and stakeholder engagement activities across the Action. It monitors gender balance and ECI participation against agreed targets, manages the Action website and digital communication channels, and produces annual gender and inclusiveness monitoring reports for the Management Committee. WG5 ensures coherent and sustained outreach to scientific, practitioner and public audiences, and plays a strategic integration role by connecting dissemination activities across all five WGs into a unified communication and impact narrative.

WG5.T1: Training Schools and capacity building programme - Organisation of at least four Training Schools over the Action period — co-located with ITC institutions in at least three cases — covering interdisciplinary urban heat research methods. All curricula and learning materials will be made freely available online in accessible formats following each event.

WG5.T2: STSM and Virtual Mobility programme management - Administration, promotion and monitoring of the STSM and Virtual Mobility programme, with explicit priority to ECI and ITC researchers, targeting at least 60 completed missions over the Action period, of which at least 60% involve ITC institutions.

WG5.T3: Policy communication and stakeholder engagement - Production of Policy Briefs in coordination with WG4, maintenance of the urban heat knowledge portal, coordination of webinars for city networks, management of digital communication channels, and annual reporting on geographic and demographic distribution of dissemination reach.

3.2 Network of Proposers

Organisation	Country	Type	Role	Description
University of Athens	Greece	University / Higher Education Institution	Coordinator	Action Chair and Coordinator; leads WG1 and overall network management
Representative institution – Romania	Romania	University	Partner	WG3 co-lead; ITC anchor institution for Eastern Europe
Representative institution – Bulgaria	Bulgaria	University	Partner	WG4 co-lead; governance analysis and ITC stakeholder engagement
Representative institution – Croatia	Croatia	Research Institute	Partner	WG2 co-lead; NbS evidence synthesis
Representative institution – Serbia	Serbia	University	Partner	WG3 contributor; health vulnerability analysis, ITC
Representative institution – Portugal	Portugal	University	Partner	WG1 contributor; climate metrics and remote sensing, ITC
Representative institution – Czech Republic	Czech Republic	Research Institute	Partner	WG2 contributor; urban greening and ecosystem services, ITC
Representative institution – Poland	Poland	University	Partner	WG4 contributor; urban planning and governance, ITC
Representative institution – Netherlands	Netherlands	University	Partner	WG2 and WG3 contributor; epidemiology and NbS co-benefits
Representative institution – Spain	Spain	University / Research Centre	Partner	WG1 and WG3 contributor; Mediterranean urban heat and health
Representative institution – Germany	Germany	Research Institute	Partner	WG4 contributor; governance and EU policy interface

URBANHEAT assembles a geographically diverse and disciplinarily complementary network deliberately structured to exceed COST eligibility requirements on ITC participation. The initial Network of Proposers spans at least twelve COST Member Countries, with ITC countries comprising the majority. ITC institutions are positioned as WG co-leads and as co-organisers of Training Schools, ensuring substantive scientific leadership rather than nominal participation. The network includes universities, research institutes, and institutions with direct organisational links to public health authorities and municipal governments, providing an organic interface to practitioner and policy stakeholder communities.

Early-career investigators are embedded in WG leadership at deputy level and are the priority beneficiaries of STSM and Virtual Mobility instruments. Gender balance in WG leadership targets a minimum of 40% representation for each gender, monitored and enforced through a dedicated Gender Balance Coordinator role within the Management Committee. This structural commitment to gender equity extends to speaker selection at Action events and co-authorship of major outputs.

The consortium's complementarity is structural and non-replicable at the national level: no single

institution possesses the full disciplinary, geographic and stakeholder portfolio that URBANHEAT assembles. The European added value of the network is therefore intrinsic — arising directly from the combination of climate science, epidemiology, urban planning, governance analysis and practitioner engagement across multiple national contexts — and constitutes the core justification for COST-level coordination.

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	Action launch and WG constitutive meetings completed	WG1-WG5	M6	Kick-off meeting report and WG membership lists submitted to Management Committee
MS2	Harmonised urban heat indicator framework agreed and published	WG1	M18	Framework document published on Action website and submitted to open-access repository
MS3	First Training School completed with ITC co-organisation	WG5	M18	Training School report including participant statistics (ECI%, ITC%, gender balance) submitted
MS4	NbS evidence synthesis report and typology published	WG2	M24	Systematic review manuscript submitted to peer-reviewed journal; pre-print available
MS5	Heat-health vulnerability maps produced for participating countries	WG3	M30	Maps and underlying data deposited in open-access FAIR-compliant repository
MS6	Comparative governance analysis and first Policy Brief series published	WG4	M30	Governance analysis report and at least two Policy Briefs publicly available on Action website
MS7	Mid-term review completed and second-phase work plan adopted	WG1-WG5	M24	Mid-term review report approved by Management Committee and submitted to COST Association
MS8	Final integrated urban heat adaptation guidance document published	WG1-WG5	M48	Guidance document publicly available; dissemination to EU institutions and city networks confirmed

3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Urban heat exposure and vulnerability indicator framework	WG1	M18	Report / Protocol Document	Public

D1.2	Pan-European urban heat exposure atlas (open dataset)	WG1	M36	Dataset	Public
D2.1	Systematic review and meta-analysis of NbS heat mitigation performance	WG2	M24	Peer-reviewed publication	Public (Open Access)
D2.2	Practitioner NbS guidance: context-sensitive typology and decision support tool	WG2	M36	Guidance document / web tool	Public
D3.1	Harmonised heat-health surveillance methodology protocol	WG3	M24	Protocol document	Public
D3.2	Cross-national analysis of heat-health outcomes and social vulnerability	WG3	M36	Peer-reviewed publication	Public (Open Access)
D4.1	Comparative urban heat governance mapping report	WG4	M30	Report	Public
D4.2	Policy Brief series on urban heat governance (minimum 8 briefs over 4 years)	WG4	M48	Policy briefs	Public
D5.1	Training School curricula and open-access learning materials (4 schools)	WG5	M48	Educational materials	Public
D5.2	Final integrated urban heat adaptation guidance for European cities	WG1-WG5	M48	Guidance document	Public

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation
R1	Coordination load on Action Chair and WG leads exceeds capacity of volunteer researchers, risking delays in deliverable production	Medium	High	Distribute coordination responsibility across WG co-leads and a dedicated Vice-Chair; designate an Action Science Communication Manager within WG5; build realistic timelines with buffer months before key

				milestones.
R2	Stakeholder and city partner continuity: municipal contacts change due to elections or administrative restructuring, disrupting co-production activities in WG4	Medium	Medium	Establish formal Memoranda of Understanding with city networks (Eurocities, Covenant of Mayors) as institutional anchors rather than relying on individual contacts; maintain a diversified portfolio of at least ten city partners.
R3	Insufficient ITC researcher participation in STSMs and Training Schools due to financial and administrative barriers at sending institutions	Medium	High	Proactively promote STSM and YRI grant opportunities in ITC countries from Month 1; co-locate Training Schools in ITC institutions; simplify administrative procedures and provide pre-application guidance via WG5.
R4	Disciplinary silos persist within the network, reducing cross-WG integration and the quality of interdisciplinary outputs	Low	High	Programme joint cross-WG workshops annually; require all major deliverables to carry co-authorship from at least three disciplines; establish a Scientific Integration Group at MC level with representatives from each WG.
R5	Open data sharing impeded by national data governance restrictions, limiting the scope of the urban heat exposure atlas	Medium	Medium	Conduct a data governance mapping exercise in Year 1 (WG1.T1); design the atlas to accommodate partial or aggregated data contributions where full sharing is not legally permissible; use federated analysis approaches where necessary.